

Database Normalization Report - Online School Management System

This report summarizes the normalization process applied to the Online School Management System database. The schema was designed using ER modeling, converted to a relational model, and normalized to improve data integrity and reduce redundancy.

First Normal Form (1NF)

All attributes are atomic. No repeating groups or multi-valued attributes exist in the schema.

Second Normal Form (2NF)

All non-key attributes are fully dependent on the primary key. Partial dependencies were eliminated through relation decomposition.

Third Normal Form (3NF)

Transitive dependencies were removed. Related information is stored in separate tables connected through foreign keys.

Boyce-Codd Normal Form (BCNF)

All identified determinants are candidate keys or superkeys. No BCNF violations were identified.

Fourth Normal Form (4NF)

No non-trivial multivalued dependencies were identified in the schema.

Fifth Normal Form (5NF)

No non-trivial join dependencies were identified. The schema does not require further decomposition.

Conclusion: The database satisfies 3NF and is consistent with BCNF, 4NF, and 5NF. The resulting design minimizes redundancy, improves consistency, and provides a scalable foundation for implementation.